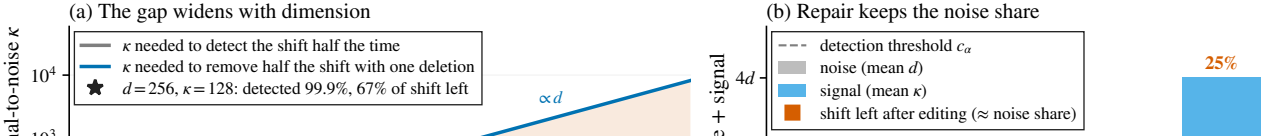
(a) The gap widens with dimension
κ needed to detect the shift half the time
κ needed to remove half the shift with one deletion
d = 256, κ = 128: detected 99.9%, 67% of shift left
∝ d
detectable, not repairable
∝ √d
Calibration signal-to-noise κ
10^4
10^3
10^2
10^1
16
64
256
1024
4096
Representation dimension d
(b) Repair keeps the noise share
detection threshold c_α
noise (mean d)
signal (mean κ)
shift left after editing (≈ noise share)
||X||^2 = noise + signal
4d
2d
d
0
84%
67%
50%
25%
detected 63%
detected 99.9%
detected >99.9%
detected >99.9%
3√d
d/2
d
3d
Calibration signal-to-noise κ (d = 256)